

# NicheJobs — Niche Remote Job Board

## Product & Build Specification (v1.1 — 20 September 2026 — multi-stream revenue plan)

**Working name:** "NicheJobs" is a placeholder. The owner will replace the name and choose the niche (see Section 2). Build the app so name, niche, prices and seed companies are **configuration, not hard-coded**.

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### 0. How to use this document (instructions for the AI model)

You are receiving a complete product specification for a web app. Your job is to **design and build it** (UI design, code, database, deployment notes).

Rules for you:

1. Read the whole document before writing any code.
2. Follow the tech stack in Section 9 unless you find a real technical problem. If you deviate, say why in one line.
3. Where the document says **DEFAULT**, use that value. The owner can change it later through config.
4. Build in the milestones from Section 21, in order. After each milestone, list what works and what is left.
5. Keep the backend **small**: no separate server, no queues, no microservices. Next.js route handlers + Postgres + a daily cron job only.
6. Do not invent features that are not in this document. Put ideas in a "Suggestions" list at the end of your answer.
7. When something is unclear, choose the simplest option, state your assumption, and continue.

### Suggested first prompt to the AI model:

"Read the attached specification. Start with Milestone 1 (project setup, database, config). Give me the full code, the Prisma schema, and step-by-step instructions to run it locally. Then wait for me to say 'next milestone!'"

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### 1. Product summary (plain English)

**What it is:** A website that lists jobs in **one narrow niche** (for example: remote cybersecurity jobs). Job seekers use it **for free**. **Employers pay** to post a job or to make a job "Featured".

**How it makes money (several streams, switched on in this order — details in Section 5):**

1. Partner job networks pay per click for the jobs they supply (backfill) and affiliate partners pay commissions — from the first day there is traffic.
2. Employers pay for job listings (about \$149 for 30 days) and for featured placement.
3. Sponsors and advertisers pay for newsletter slots.
4. Repeat employers pay a monthly plan.
5. Later: recruiter database access, a paid candidate tool, and career-fair events.

**Why it is possible for one developer:**

- The website is mostly pages and forms. No heavy backend.
- The board is **never empty**: a daily job pulls open jobs from companies' own public career systems (Greenhouse, Lever, Ashby) and lists them with a link to the employer's apply page.
- Google Jobs sends free traffic when each job page has correct structured data.

**How money is earned, in one line:** Traffic (from Google + the newsletter) is the asset.

Different buyers pay to reach it: employers (listings, featured, plans), partner job networks (per click), sponsors (newsletter slots), and affiliate partners (commission per sale). Being "famous" does not pay by itself — each stream has a specific payer.

**Who it serves worldwide:** Candidates from any country (free). Employers from any country pay in USD (or local currency through the payment provider).

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## 2. Decisions and defaults

| Topic                   | DEFAULT (use this unless the owner changes it)                                                                                                                                                                                                            |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Site name               | <code>NicheJobs</code> (placeholder — read from <code>config/site.ts</code> )                                                                                                                                                                             |
| Niche                   | <b>Chosen by owner.</b> Examples in this document use "remote cybersecurity jobs". All niche words come from <code>config/niche.ts</code> (keywords, categories, filters)                                                                                 |
| Language / currency     | English. Prices in USD                                                                                                                                                                                                                                    |
| Candidate accounts      | <b>None.</b> Candidates never log in. Optional email for job alerts only                                                                                                                                                                                  |
| Employer accounts       | <b>None.</b> Employers use a secret manage link sent by email (and optional company claim, Flow E)                                                                                                                                                        |
| Job duration            | 30 days from publish date                                                                                                                                                                                                                                 |
| Price: Standard listing | \$149 (config value)                                                                                                                                                                                                                                      |
| Price: Featured listing | \$249 (config value)                                                                                                                                                                                                                                      |
| Moderation              | First listing of a new employer email is reviewed by the owner. After one approved listing, that employer is auto-approved                                                                                                                                |
| Payment provider        | Provider interface with adapters for <b>Stripe Checkout</b> and <b>Lemon Squeezy</b> . Chosen by env var <code>PAYMENT_PROVIDER</code> . (Owner picks based on which provider accepts his country/business. Implement Stripe first, Lemon Squeezy second) |
| Hosting                 | Vercel (or any Node host). Database: Postgres (Neon or Supabase free tier)                                                                                                                                                                                |
| Email                   | Resend (transactional + newsletter)                                                                                                                                                                                                                       |
| Search                  | Postgres full-text search (no external search service)                                                                                                                                                                                                    |
| Analytics               | Plausible or Umami (privacy friendly) + own <code>JobClick</code> table for apply clicks                                                                                                                                                                  |
| Extra revenue at launch | Partner job feed backfill (pay per click), affiliate offer blocks, and a sponsor page ( <code>/advertise</code> ). Banner ads later                                                                                                                       |

| Topic                | DEFAULT (use this unless the owner changes it)                                                                         |
|----------------------|------------------------------------------------------------------------------------------------------------------------|
| Partner job networks | Owner applies to 2–3 networks (approval and traffic requirements vary). Until approved, the feature stays switched off |

Prices above are **starting values to test**, not proven prices. Store them in config so the owner can change them without code changes.

### 3. Who uses the app (actors)

| Actor                         | Description                                                                           | Logs in?                                          | Pays?      |
|-------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------|------------|
| <b>Candidate</b>              | Job seeker anywhere in the world. Finds jobs, clicks Apply, may subscribe to alerts   | No                                                | No (free)  |
| <b>Employer</b>               | Hiring manager, recruiter, or founder who wants to reach niche candidates             | No (secret link; optional verified company claim) | <b>Yes</b> |
| <b>Owner / Admin</b>          | The founder. Approves listings, adds company sources, sends newsletter, does outreach | Yes (email allow-list)                            | —          |
| <b>Google (search engine)</b> | Crawls job pages, shows them in Google Jobs                                           | —                                                 | —          |
| <b>Cron / System</b>          | Daily job import, expiry, reminders, newsletter drafts                                | —                                                 | —          |

### 4. Why people use it (value for each side)

**Candidate (free):** One place with only relevant jobs. Every job has a clear apply link to the employer. Salary and remote rules are shown whenever the company published them. Email alerts for new jobs.

**Employer (pays):**

- Reach a **targeted** audience instead of paying for a huge general board.
- Appear in **Google Jobs** and in the weekly newsletter.
- **Featured** listing stays at the top of the list for 30 days.
- A simple stats page: how many people viewed the job and clicked Apply. This proves the value and makes renewal easy.
- 5-minute posting, no account, no subscription needed.

**Owner:** Income from several payers at once (see Section 5): partner networks, affiliates, employers, and sponsors. Imported jobs fill the site so it has traffic; the traffic is what gets sold.

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## 5. Business model — who benefits and how the owner earns money

### 5.1 In plain English: who benefits, who pays

The site is useful for candidates, but **candidates usually do not pay** (industry guides say over 95% of job boards earn from employers, because candidates rarely pay when free options exist). The owner earns by selling access to the site's audience to several different buyers.

**Traffic alone does not pay. It pays when a specific buyer wants to reach that traffic.**

| Who                                                                                  | What they get                                                | Do they pay?                                 |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------|
| Candidate                                                                            | Relevant jobs in one place, free email alerts                | No (an optional paid tool may come later)    |
| Employer                                                                             | Reach to a targeted audience, Google Jobs, newsletter, stats | <b>Yes</b> — listing, featured, monthly plan |
| Partner job network                                                                  | Clicks on the jobs it distributes                            | <b>Yes</b> — per click or per application    |
| Sponsor / advertiser (bootcamps, tool vendors, certification providers, conferences) | Reach to the niche audience                                  | <b>Yes</b> — newsletter slot, banner         |
| Affiliate partner (resume services, courses, certification vouchers)                 | New customers                                                | <b>Yes</b> — commission per sale             |
| Owner                                                                                | Income from all of the above                                 | —                                            |

### 5.2 Revenue streams

Benchmarks come from industry guides (2026) and are **starting assumptions to test in the chosen niche**, not promises.

| # | Stream                                                     | Who pays                                                                      | Benchmark / default                                                                                                                                                                  | Effort                                      | Start when                                                          |
|---|------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------|
| 1 | <b>Partner-feed backfill</b> (pay per click / application) | Job networks (for example Jooble, Adzuna, Appcast Clickcast; approval needed) | About \$0.10–\$0.50 per click for small boards                                                                                                                                       | Low: feed import + click tracking (Flow J)  | At launch, as soon as a network approves the site                   |
| 2 | <b>Affiliate offers</b>                                    | Resume services, courses, certification providers, tools                      | Commission 10–30% per sale                                                                                                                                                           | Low: offer blocks + click tracking (Flow L) | At launch                                                           |
| 3 | <b>Paid job listing</b> (Standard, 30 days)                | Employers                                                                     | Default \$149. Niche boards charge roughly \$100–\$600 depending on how hard the audience is to reach; many operators underprice at first                                            | Medium: outreach                            | Month 1–2                                                           |
| 4 | <b>Featured upgrade</b> (top of list + newsletter slot)    | Employers                                                                     | Default \$99 for an imported job; \$249 bundle for a new direct listing. Pin-to-top upgrades commonly sell for about \$29–\$100 and are bought by roughly 15–25% of paying employers | Low                                         | Month 1–2                                                           |
| 5 | <b>Newsletter sponsor slot</b>                             | Tool vendors, bootcamps, conferences                                          | About \$200–\$500 per slot at about 2,000 subscribers. Default \$250                                                                                                                 | Low–Medium                                  | When the newsletter has about 1,000+ confirmed subscribers (Flow K) |
| 6 | <b>Employer monthly plan</b>                               | Repeat hirers, staffing agencies                                              | About \$299–\$499/month. Default \$399                                                                                                                                               | Low (billing)                               | After about 20 employers have paid for listings                     |
| 7 | <b>Later options</b>                                       | Various                                                                       | Recruiter resume database \$99–\$500/month (needs 500 to                                                                                                                             | High                                        | Only after real audience size                                       |

| # | Stream | Who pays | Benchmark / default                                                                                                                                                                                             | Effort | Start when |
|---|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------|
|   |        |          | several thousand candidate profiles); paid candidate AI tools (about \$9/month benchmark); banner ads (AdSense about \$2-8 CPM; directly sold about \$5-15 CPM); virtual career fairs (booths from about \$250) |        | (not MVP)  |

### 5.3 The order to switch things on (money ladder)

1. **Traffic stage (months 0-3):** fill the board (ATS import + partner feed), SEO pages, newsletter signups. Earn from **streams 1 and 2** while traffic grows.
2. **Proof stage (months 1-6):** show employers real views and apply clicks; sell **streams 3 and 4**.
3. **Recurring stage (months 4-12):** add **streams 5 and 6**.
4. **Scale stage (12+ months):** consider **stream 7** only if the audience is large.

Rules: do not start with subscriptions or a resume database (there is nobody to sell to yet). Do not depend on one stream: guides report that about half of boards earn 81-100% of revenue from listings alone, and boards that stay at a single pricing model tend to plateau at roughly \$3,000-\$5,000/month, while boards that layer several streams grow past that.

### 5.4 Pricing defaults (all editable in `config/pricing.ts`)

| Item                                                   | DEFAULT          |
|--------------------------------------------------------|------------------|
| Standard listing, 30 days                              | \$149            |
| Featured listing (new direct listing), 30 days         | \$249            |
| Feature upgrade for an imported job, 30 days           | \$99             |
| Renewal                                                | Same as original |
| Employer monthly plan (after MVP)                      | \$399/month      |
| Newsletter sponsor slot (invoiced by the owner in MVP) | \$250 per issue  |
| Test higher prices after the first 10 paid listings.   |                  |

## 5.5 Refund policy (DEFAULT)

- Full refund if the owner rejects the listing or cannot publish it.
- No refund after publish, except at the owner's discretion.

## 5.6 Example: where the money goes for one listing (merchant-of-record provider)

Employer pays **\$149.00**. Provider fee example:  $5\% + \$0.50 = \$7.95$ . Net to owner: **\$141.05** (paid out to the owner's bank on the provider's payout schedule). Other providers have different fees. The app must store the gross amount, and the provider fee if the provider reports it.

## 5.7 Example: where the money goes for one partner click

Priya clicks Apply on a partner job. The click is logged and she is redirected to the partner's tracking link. The partner counts the click and pays the owner monthly according to the publisher agreement, for example \$0.25. **1,000 such clicks = about \$250**. Each click is small, so this stream needs traffic volume. It is useful because it starts earning before any employer has paid.

## 5.8 Illustrative monthly income (NOT a forecast)

Assumptions: partner-feed clicks average \$0.25; a share of visitors click partner jobs (15% / 20% / 25%); Standard listings \$149; feature upgrades \$99; sponsor slots \$250; monthly plans \$399.

| Scenario                           | Visitors<br>/ month | Partner<br>clicks | Partner<br>income | Direct<br>listings                | Feature<br>upgrades        | Sponsors                    | Plans                             | Affiliate | Total          |
|------------------------------------|---------------------|-------------------|-------------------|-----------------------------------|----------------------------|-----------------------------|-----------------------------------|-----------|----------------|
| Slow<br>(about<br>months<br>3-6)   | 5,000               | 750               | \$188             | $3 \times$<br>\$149 =<br>\$447    | —                          | —                           | —                                 | —         | <b>\$635</b>   |
| Base<br>(about<br>months<br>9-12)  | 20,000              | 4,000             | \$1,000           | $10 \times$<br>\$149 =<br>\$1,490 | $5 \times \$99 =$<br>\$495 | $2 \times \$250$<br>= \$500 | —                                 | \$200     | <b>\$3,685</b> |
| Good<br>(about<br>months<br>15-18) | 30,000              | 7,500             | \$1,875           | $18 \times$<br>\$149 =<br>\$2,682 | $8 \times \$99 =$<br>\$792 | $3 \times \$250$<br>= \$750 | $2 \times$<br>\$399<br>=<br>\$798 | \$250     | <b>\$7,147</b> |

For comparison, one industry guide says a well-executed niche board can reach about \$3,000–\$7,000/month blended at about 18 months. Many boards reach less. Results depend on niche,

traffic and outreach.

## 5.9 Honest limits

- Partner networks may reject a small or new site or set traffic minimums. Some large publisher programs no longer accept new publishers. Apply early and expect to wait.
  - Per-click income is low; treat it as the "bridge" income while employer sales start.
  - Growth of traffic matters more than early revenue. A common guide target is 20–30% month-over-month visitor growth in the early months.
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## 6. End-to-end flows (with worked examples)

Fictional people used in examples:

- **Priya** — candidate looking for a remote SOC analyst job.
- **Maria Lopez** — Head of Security at **Acme Cloud**, wants to hire.
- **Northwind Security** — a company that already posts jobs on Greenhouse.
- **The Owner** — the founder running NicheJobs.

### Flow A — Candidate finds a job from Google and applies (free traffic → value)

1. Priya types "remote SOC analyst jobs" in Google. Google Jobs shows a result from NicheJobs (works because every job page has `JobPosting` structured data — Section 12).
2. She clicks it and lands on `/jobs/soc-analyst-acme-cloud-x7k2`.
3. The page shows: job title, company name and logo, "Remote — eligible countries: India, Philippines, Poland...", salary range (if published), posted date, full description, a **"Apply on company site"** button, a small company card, 5 similar jobs, and a box "Get new jobs like this by email".
4. She clicks **Apply**. The link goes to `/api/click/{jobId}`. The server saves one row in `JobClick` (jobId, time, referrer) and redirects (HTTP 302) to the employer's apply URL with `?utm_source=nichejobs` added.
5. She finishes the application on the employer's site. **NicheJobs never sees her CV.**
6. Optional: before leaving, she enters her email and picks "SOC / detection roles" and "Remote worldwide". She receives a confirmation email (double opt-in). After she clicks the confirm link, she is subscribed.

**What the system stores:** page view (analytics), one apply click, optional confirmed subscriber.

**Why this matters for revenue:** apply clicks are the proof shown to employers (Flow C step 9 and Flow E).

## Flow B — Candidate browses and uses filters

1. Home page (/) shows a search box and the newest jobs. Featured jobs are pinned at the top with a highlight.
2. Filters: category, seniority, employment type, remote region/country, salary present, posted date (24h / 7d / 30d).
3. Search uses Postgres full-text search on title, company, description.
4. URL reflects filters (/?q=soc&region=india&type=full-time) so results can be shared.
5. Static-looking landing pages exist for common searches, e.g. /remote-soc-analyst-jobs, /remote-jobs-in-india (Section 8).
6. **Job alerts:** Subscribers with saved filters receive a daily or weekly email listing only new matching jobs (max 20). Every alert has a one-click unsubscribe.

## Flow C — Employer posts a job and pays (MAIN REVENUE FLOW)

Example: Maria Lopez wants to hire a SOC analyst for Acme Cloud.

1. **Discovery.** Maria sees the board through a newsletter, a link in the Owner's outreach email, or Google. She clicks "**Post a job**" in the header → /post-a-job.
2. **Step 1 — Job details.** She fills:
  - Job title\* ("SOC Analyst (Remote)")
  - Category\* (from niche config)
  - Employment type\* (full-time / part-time / contract)
  - Location type\* (Remote / Hybrid / On-site)
  - If Remote: eligible countries or regions\* (multi-select; "Worldwide" allowed)
  - Salary range (optional; currency, min, max, per year/month/hour) — the form encourages it ("Jobs with salary get more clicks")
  - Description\* (rich text or markdown, min 300 characters)
  - How to apply\* (an apply URL or an email address)
3. **Step 2 — Company.** Company name\*, website\*, logo (upload, max 200 KB, png/jpg/svg/webp), one-line company description, and her **work email\*** (used for receipts and the manage link).
  - If the email is a free provider (gmail, yahoo...), the app **flags the listing for manual review** (not blocked).
4. **Step 3 — Choose plan.** Cards: **Standard \$149** and **Featured \$249**. A live preview shows exactly how the job will look in the list and on its page. She ticks "I agree to the Terms".
5. **Pay.** Clicking "Pay \$149" creates an **Order** (status **PENDING**) and a **Job** (status **PENDING\_PAYMENT**) and redirects her to the payment provider's hosted checkout page. She pays by card / wallet.

6. **Payment confirmed.** The provider calls `POST /api/webhooks/payments`. The app verifies the signature, marks the `Order` `PAID`, and moves the `Job`:
  - to `PENDING_REVIEW` if this employer email is new (Owner must approve), or
  - straight to `LIVE` if the employer email already has an approved listing.
7. **Emails to Maria:** (a) provider receipt; (b) "We received your listing" with her **manage link** `/manage/{editToken}` (a long random secret token).
8. **Owner approval** (first-time employers). The Owner opens `/admin/review`, sees the job, company website, email domain and warnings, and clicks **Approve** or **Reject (with reason)**.
  - Approve → status `LIVE`, `publishedAt = now`, `expiresAt = now + 30 days`, page revalidated, URL sent to Google Indexing API, sitemap updated, email "Your job is live" to Maria with the public link.
  - Reject → email with reason and **automatic refund request** to the payment provider; `Order` becomes `REFUNDED`.
9. **Manage page** `/manage/{editToken}`: shows views and apply clicks per day, buttons **Edit text**, **Close job**, **Upgrade to Featured** (pay the difference), **Renew**, **Download invoice link**.
10. **Expiry.** On day 25 the system emails "Your listing expires in 5 days — renew?". On day 30 the job becomes `EXPIRED`: the page shows an "expired" banner, is set to `noindex`, the `JobPosting` markup is removed, and the URL is sent to Google as removed (`URL_DELETED`). Similar jobs are shown so the page stays useful.

**Result for the Owner:** \$149 gross, about \$141 net (example provider fees), with about 2 minutes of review work for a new employer.

### Flow D — Automatic job import (keeps the board full)

Example: Northwind Security uses Greenhouse with board token `northwindsecurity`.

1. The Owner opens `/admin/sources` and clicks **Add source**: provider = Greenhouse, board token = `northwindsecurity`, company name and website.
2. Every night at 02:00 UTC, the cron endpoint `/api/cron/sync-ats` (protected by a secret header) runs. For every active source it:
  1. Fetches the public job feed of that provider.
  2. Normalizes each job to the internal format (Section 11).
  3. Keeps only jobs that match the niche keywords and categories in `config/niche.ts`.
  4. Upserts the job by `(source, sourceJobId)`. New jobs become `LIVE` with `source = ATS` and the label "**Direct from company career page**".
  5. Any job that was not seen in **2 consecutive runs** is set to `EXPIRED`.

- Writes a `SyncLog` row (jobs seen, added, updated, expired, errors).
- Example: Northwind has 24 open jobs. 3 match the niche ("SOC Analyst", "Detection Engineer", "Security Engineer"). Those 3 appear on the board that night, each with an apply button that goes to Northwind's own Greenhouse page.
- If a source fails 3 nights in a row, the Owner receives an alert email.
- Imported jobs are **read-only** (the employer's career system is the source of truth). Only the "Featured" flag can be bought (Flow E).

### Flow E — Owner reaches out, employer claims company and upgrades (SECOND REVENUE FLOW)

- In `/admin/companies` the Owner sees Northwind Security: 3 imported jobs, **1,240 views and 96 apply clicks in the last 30 days**.
- The Owner sends a short manual email to the company's hiring contact (template in Section 19): "Your 3 security roles are already listed on NicheJobs and got 96 apply clicks this month. Want them featured at the top for \$99 each?"
- The contact clicks the link `/claim/northwind-security`.
- Claim verification:** they enter a work email. The email domain must match the company website domain (`@northwindsecurity.com`). The app sends a **magic link**. Clicking it creates a `CompanyClaim` with `verifiedAt`.
- Company dashboard** `/company/{companySlug}/dashboard` (opened via the magic link session): stats for every job, buttons **Feature this job (\$99)**, **Edit company profile** (logo, description, website), **Post a new job**.
- Clicking **Feature this job** starts checkout (same payment flow as Flow C). After payment (webhook) `isFeatured = true`, `featuredUntil = now + 30 days`. The job jumps to the top of its lists and is included in the next newsletter.

### Flow F — Weekly newsletter

- Every Tuesday 06:00 UTC, the system creates a **draft issue** in `/admin/newsletter`: featured jobs first, then the 10 best new jobs of the week, plus an optional sponsor block.
- The Owner previews it, edits the intro sentence, and clicks **Send**.
- Only **confirmed** subscribers receive it. Every email has an unsubscribe link and the physical/contact address required by law.
- Links carry `utm_source=newsletter` so clicks are counted.

### Flow G — Owner moderation and daily operations

Owner's daily routine (about 15 minutes):

1. `/admin` dashboard: revenue this month, paid listings, live jobs, sync health, visitors, apply clicks, newsletter subscribers.
2. `/admin/review`: approve or reject pending listings. Checklist shown next to each job: company website loads? apply link works? email domain matches website? scam signals (asks candidates to pay, crypto payments, "no experience, \$5k/day")? If any red flag → reject.
3. `/admin/sources`: check failing sources, add new company sources.
4. `/admin/takedowns`: handle removal requests from the public form `/takedown` (company asks to remove an imported job). Owner sets status `REMOVED` and the source is blocklisted. Target response: within 48 hours.

## Flow H — Payment lifecycle and edge cases

| Event                      | System behaviour                                                                                                                                                                                                 |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Checkout completed         | Webhook → <code>Order = PAID</code> , job moves forward (Flow C step 6)                                                                                                                                          |
| Payment failed / abandoned | <code>Order = FAILED</code> or stays <code>PENDING</code> . Job stays <code>PENDING_PAYMENT</code> . Reminder email after 1 hour and after 24 hours ("Finish your listing"). Unpaid drafts deleted after 14 days |
| Duplicate webhook          | Ignored (store <code>providerEventId</code> , unique)                                                                                                                                                            |
| Refund                     | Webhook or admin action → <code>Order = REFUNDED</code> , job <code>REMOVED</code> , employer emailed                                                                                                            |
| Wrong amount / currency    | Job not published, Owner alerted                                                                                                                                                                                 |
| Provider outage            | Checkout button shows a friendly error; draft is saved                                                                                                                                                           |

## Flow I — Renewal and repeat business

- Day 25: renewal reminder with one-click "Renew for \$149".
- Renewal creates a new `Order` and resets `expiresAt` (+30 days). The URL stays the same.
- After an employer has 3 paid listings in 6 months, the app shows a banner on the manage page: "Save with a monthly plan" (after-MVP feature; collect interest by email for now).

## Flow J — Partner job feed: earning per click (backfill income)

Example: the Owner has been approved by a partner job network (called "JobNet" here). The network gives a feed URL and pays per click.

1. The Owner opens `/admin/partners` and adds **JobNet**: feed URL, active = on, estimated payout per click (for reports only).
2. Every night `/api/cron/sync-feeds` (secret header) downloads the feed and, for each job: applies the **same niche keyword filter** as the ATS import, sanitizes text, and upserts it with `origin = FEED`, status `LIVE`. The job's `applyUrl` stores the partner's tracking link **unchanged**. A job missing from the feed for 2 runs becomes `EXPIRED`. Max jobs per partner is limited by config (DEFAULT 500).
3. On the site, partner jobs appear in a separate section "**More jobs from our partners**" below the direct and ATS jobs, each marked **Sponsored**. They are never mixed above paid or direct jobs. Partner job pages are `noindex`, have no JobPosting markup, and are not in the sitemap.
4. Priya clicks Apply on a partner job → `/api/click/{jobId}` saves a click row → redirects to the partner tracking link.
5. The network counts the click and pays the Owner monthly under the publisher agreement (paid outside the app). In `/admin/partners` the Owner sees clicks per partner and **estimated** income (clicks × estimated payout) to compare with the real payout.

Rules: follow each partner's publisher terms (attribution, freshness, no changing links); label partner jobs clearly; if a partner asks for removal, switch it off.

### Flow K — A sponsor buys a newsletter slot

Example: **CyberLearn Academy** (a fictional bootcamp) wants to reach security job seekers.

1. Their marketing manager finds `/advertise`. The page shows the audience (confirmed subscribers, monthly visitors), the price (\$250 per issue, from config), and what is included (logo, 40-word text, one link).
2. They fill the inquiry form: company, work email, link, headline, text, logo. The app creates a `SponsorBooking` with status `INQUIRY` and emails the Owner.
3. The Owner replies with an invoice or a payment link from the payment provider's dashboard (**no in-app checkout in MVP**). When paid, the Owner marks the booking `PAID` and picks the issue date in `/admin/sponsors`.
4. The next newsletter draft (Flow F) automatically includes the sponsor block, clearly labeled **Sponsored**. After sending, the booking becomes `SENT` and the Owner can email the sponsor the click count.

### Flow L — Affiliate offers for candidates

Example: a resume-review service and a certification-exam voucher provider have affiliate programs.

1. The Owner adds offers in `/admin/affiliates`: name, affiliate link, short headline and text, category (for example RESUME, COURSE, CERTIFICATION, TOOL), where to show it (JOB\_PAGE, ALERT\_EMAIL, NEWSLETTER), active on/off.
2. On a job page, a small box in the sidebar shows one relevant offer ("Preparing your resume for this role? {offer}") with the note "We may earn a commission if you buy."
3. Priya clicks → `/api/aff/{offerId}?placement=JOB_PAGE` logs an `AffiliateClick` and redirects to the affiliate link. The link uses `rel="sponsored nofollow"`.
4. If she buys, the partner's dashboard shows the sale and pays the commission (10-30% is typical). The Owner records commissions in the admin page for reports.

Rules: at most one affiliate block per page/email; never inside the job description; always show the disclosure; offers must be relevant and honest (no fake claims).

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## 7. Features and acceptance criteria

### 7.1 MVP (must be built)

| ID  | Feature                               | Acceptance criteria (how to know it is done)                                                                                            |
|-----|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| F1  | Job list + search + filters           | List shows LIVE jobs, featured first, then newest. Filters combine (AND). Pagination 20 per page. Empty state has a helpful message     |
| F2  | Job detail page                       | Shows all fields, apply button, similar jobs, alert signup. Has valid <code>JobPosting</code> JSON-LD (passes Google Rich Results Test) |
| F3  | Apply click tracking                  | Apply button → <code>/api/click/{id}</code> → logs → 302 to apply URL. Direct link is also present in HTML for crawlers                 |
| F4  | Company pages                         | <code>/companies/{slug}</code> shows logo, description, website, and its live jobs                                                      |
| F5  | Post-a-job (3 steps) + preview        | Form validation, draft saved on server, preview identical to real page                                                                  |
| F6  | Checkout + webhook                    | Paid order always leads to correct job status; duplicate webhooks harmless                                                              |
| F7  | Employer manage page                  | Secret link, edit, close, renew, upgrade, stats                                                                                         |
| F8  | Owner review queue                    | Approve/reject with emails and refund trigger                                                                                           |
| F9  | ATS import (Greenhouse, Lever, Ashby) | Nightly cron; niche filter; dedupe; expiry after 2 missed runs; SyncLog                                                                 |
| F10 | Company claim + feature upgrade       | Domain-verified magic link; buy Featured for imported jobs                                                                              |
| F11 | Email alerts + double opt-in          | Confirm link, daily/weekly alert, unsubscribe in one click                                                                              |
| F12 | Weekly newsletter draft + send        | Draft auto-created; Owner edits and sends                                                                                               |
| F13 | SEO                                   | Sitemap, robots, canonical URLs, JobPosting JSON-LD, Google Indexing API pings, programmatic list pages                                 |
| F14 | Admin dashboard                       | Revenue, jobs, sources, sync health, takedowns                                                                                          |

| ID  | Feature                                   | Acceptance criteria (how to know it is done)                                                                                                                                                                        |
|-----|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F15 | Legal pages                               | Terms, Privacy, Cookie notice (only if needed), Takedown form, Contact                                                                                                                                              |
| F16 | Expiry + reminders                        | Day-25 reminder, day-30 expiry with markup removal and Indexing API deletion                                                                                                                                        |
| F17 | Partner job feed backfill (pay per click) | Admin adds partners; nightly import with niche filter; partner jobs shown in a separate "Sponsored" section; <code>noindex</code> ; click redirect uses partner link unchanged; clicks per partner visible in admin |
| F18 | Affiliate offer blocks                    | Admin CRUD for offers; one block on job page, alert email and newsletter; disclosure text; click logging via <code>/api/aff/{id}</code>                                                                             |
| F19 | Advertise page + sponsor bookings         | <code>/advertise</code> with stats and inquiry form; admin list with statuses; sponsor block auto-added to the newsletter draft, labeled "Sponsored"                                                                |

## 7.2 After MVP (do NOT build now)

Employer monthly plans (start after ~20 paying employers), in-app checkout for sponsors, banner ads (AdSense or directly sold), candidate profiles/resume database for recruiters, paid candidate AI tools, virtual career fairs, employer team accounts, multi-niche network, public API/RSS, mobile app.

## 8. Pages and routes

### 8.1 Public pages

| Route                                                                                        | Purpose                                | Notes                                                                                           |
|----------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------|
| <code>/</code>                                                                               | Home: search, filters, job list        | SSR/ISR. Featured jobs first                                                                    |
| <code>/jobs/[slug]</code>                                                                    | Job detail                             | <code>JobPosting</code> JSON-LD <b>only here</b> .<br>Slug = <code>title-company-shortid</code> |
| <code>/companies/[slug]</code>                                                               | Company page                           | Lists that company's LIVE jobs                                                                  |
| <code>/remote-[category]-jobs</code>                                                         | Programmatic landing page per category | Static + ISR (revalidate 1 hour). <b>No</b> JobPosting markup on list pages                     |
| <code>/remote-jobs-in-[country]</code>                                                       | Landing page per country/region        | Same rule                                                                                       |
| <code>/post-a-job</code>                                                                     | 3-step posting flow                    | Noindex not needed, but no duplicate content                                                    |
| <code>/employers</code>                                                                      | Pricing and why post here              | Shows plans, sample stats, FAQ                                                                  |
| <code>/advertise</code>                                                                      | Sponsor and advertiser page            | Audience stats, sponsor price, inquiry form (Flow K)                                            |
| <code>/subscribe</code>                                                                      | Email alert signup                     | Also embedded in job pages                                                                      |
| <code>/confirm/[token]</code> ,<br><code>/unsubscribe/[token]</code>                         | Email actions                          | One click, no login                                                                             |
| <code>/takedown</code>                                                                       | Company asks to remove imported job    | Form → admin queue                                                                              |
| <code>/terms</code> , <code>/privacy</code> , <code>/about</code> ,<br><code>/contact</code> | Legal and trust                        |                                                                                                 |
| <code>/sitemap.xml</code> , <code>/robots.txt</code>                                         | SEO                                    | Sitemap split if > 5,000 URLs                                                                   |

## 8.2 Employer pages

| Route                                                          | Purpose                                               |
|----------------------------------------------------------------|-------------------------------------------------------|
| <code>/manage/[editToken]</code>                               | Employer job management (stats, edit, renew, upgrade) |
| <code>/claim/[companySlug]</code>                              | Start company claim (email domain check + magic link) |
| <code>/company/[companySlug]/dashboard</code>                  | Verified company dashboard (after claim)              |
| <code>/checkout/success</code> , <code>/checkout/cancel</code> | After payment return pages                            |

## 8.3 Admin pages (owner only)

`/admin`, `/admin/review`, `/admin/jobs`, `/admin/companies`, `/admin/sources`, `/admin/orders`, `/admin/newsletter`, `/admin/takedowns`, `/admin/settings`.

## 8.4 Wireframes (text)

Home `/`

+-----+

NicheJobs

JobsEmployersNewsletter

[Post a job]

+-----+

Remote cybersecurity jobs, direct from companies

[ Search title, skill, company... ] [Search]

+-----+

FILTERS

Category [v]

Region [v]

Type [v]

Seniority [v]

Salary shown[]

Posted [v]

\* FEATURED

[logo] SOC Analyst Acme Cloud

Remote - India, PH \$40-60k 2d

-----

[logo] Detection Engineer Northwind

Remote - Worldwide \$70-95k 3d Direct

-----

[logo] Security Engineer ...

[ 1 2 3 ... Next ]

+-----+

Get new jobs by email: [ email ] [Subscribe]

+-----+

Job page `/jobs/[slug]`

< Back to jobs

[logo] SOC Analyst (Remote) [ Apply on company site ]

Acme Cloud - Remote (India, Philippines, Poland) - Full-time

\$40,000 - \$60,000 / year - Posted 2 days ago - Expires in 28 days

-----

|                      |                               |
|----------------------|-------------------------------|
| Description ...      | About Acme Cloud              |
| Responsibilities ... | Website, short description    |
| Requirements ...     | 3 other open jobs             |
|                      | [ Get similar jobs by email ] |

-----

Similar jobs (5)

## Post a job (3 steps)

Step 1 Job details -> Step 2 Company -> Step 3 Plan & pay

|                 |                         |
|-----------------|-------------------------|
| +-----+ +-----+ |                         |
| form fields     | LIVE PREVIEW of the job |
|                 | card and job page       |
| [Back]          | [Continue]              |
| +-----+ +-----+ |                         |

Step 3: (o) Standard \$149 ( ) Featured \$249 [ Pay \$149 ]

## Employer manage /manage/[token]

SOC Analyst (Remote) Status: LIVE Expires: 12 Nov 2026

Views: 842 Apply clicks: 61 [Edit] [Close] [Renew] [Upgrade to Featured]

Chart: views and apply clicks per day (last 30 days)

Receipt / invoice link

## 9. Architecture and tech stack

### 9.1 Stack (DEFAULT)

- **Framework:** Next.js (App Router) with TypeScript, React, Tailwind CSS. Use shadcn/ui components if helpful.
- **Database:** PostgreSQL (Neon or Supabase free tier) with **Prisma** ORM.
- **Backend:** Next.js route handlers only (app/api/\*\*). No separate server.
- **Payments:** Provider interface + Stripe Checkout adapter + Lemon Squeezy adapter (Section 13).
- **Email:** Resend (transactional + newsletter). React Email for templates.

- **File upload (logos):** Vercel Blob or any S3-compatible bucket. Max 200 KB.
- **Cron:** Vercel Cron (or GitHub Actions calling the protected endpoints).
- **Spam protection:** Cloudflare Turnstile on public forms + rate limiting.
- **Analytics:** Plausible or Umami.
- **Hosting:** Vercel (a free plan may not allow commercial use — budget for a paid plan or use another host such as Cloudflare Pages/Netlify; check current terms).

## 9.2 High-level picture

```

Candidate —> Next.js pages (ISR/SSR) —> Postgres
Employer  —> Post form —> Checkout (Stripe / Lemon Squeezy)
                        | webhook
                        ▼
                        /api/webhooks/payments —> Order PAID —> Job status
Cron (nightly) —> /api/cron/sync-ats —> Greenhouse / Lever / Ashby public
feeds
Cron (daily)   —> /api/cron/expire-and-remind —> expiry, reminders, Indexing
API
Cron (weekly) —> /api/cron/newsletter-draft
Google Indexing API <— job created / updated / removed
Resend <— transactional + newsletter emails

```

## 9.3 Rendering and caching

- Job detail and company pages: ISR (`(revalidate = 3600)`) **plus on-demand revalidation** (`(revalidatePath)`) whenever a job changes.
- Home and filtered lists: server-rendered with short cache (60 s) or `(unstable_cache)` tags per query.
- Programmatic landing pages: ISR (`(revalidate = 3600)`).
- Never expose employer email, edit tokens or admin data in public responses.

## 9.4 Environment variables

```
DATABASE_URL=
NEXT_PUBLIC_SITE_URL=
PAYMENT_PROVIDER=stripe|lemonsqueezy
STRIPE_SECRET_KEY= / STRIPE_WEBHOOK_SECRET=
LEMONSQUEEZY_API_KEY= / LEMONSQUEEZY_STORE_ID= / LEMONSQUEEZY_WEBHOOK_SECRET=
RESEND_API_KEY= / EMAIL_FROM=
CRON_SECRET=
ADMIN_EMAILS=owner@example.com
GOOGLE_INDEXING_SERVICE_ACCOUNT_JSON= (base64)
BLOB_READ_WRITE_TOKEN=
TURNSTILE_SITE_KEY= / TURNSTILE_SECRET_KEY=
```

## 9.5 Suggested folder structure

```
app/                (pages + api routes)
components/         (JobCard, FilterBar, PostJobForm, ...)
config/             site.ts niche.ts pricing.ts
lib/               db.ts payments/ email/ ats/ seo/ indexing.ts auth.ts
prisma/            schema.prisma seed.ts
emails/            React Email templates
```

## 9.6 Config files (examples)

```
ts
```

```
// config/pricing.ts
export const PRICING = {
  currency: "USD",
  standard: { cents: 14900, days: 30 },
  featured: { cents: 24900, days: 30 },
  featureUpgrade: { cents: 9900, days: 30 }, // upgrade an imported job
  monthlyPlan: { cents: 39900 }, // after MVP, when 20+ employers have
  sponsorSlot: { cents: 25000 } // per newsletter issue (invoiced by
};

// config/niche.ts (example niche: remote cybersecurity jobs)
export const NICHE = {
  name: "Remote Cybersecurity",
  categories: ["SOC / Detection", "AppSec", "Cloud Security", "GRC / Compliance",
  includeKeywords: ["security", "soc", "siem", "pentest", "appsec", "grc", "detect
  excludeKeywords: ["security guard", "physical security"],
  seniorityLevels: ["Junior", "Mid", "Senior", "Lead"]
};
```

## 10. Data model (Prisma schema — starting point)

prisma

```
generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

enum JobOrigin {
  DIRECT
  ATS
  FEED
}

enum AtsProvider {
  GREENHOUSE
  LEVER
  ASHBY
}

enum JobStatus {
  PENDING_PAYMENT
  PENDING_REVIEW
  LIVE
  EXPIRED
  REJECTED
  REMOVED
}

enum LocationType {
  REMOTE
  HYBRID
  ONSITE
}

enum EmploymentType {
  FULL_TIME
  PART_TIME
  CONTRACT
  INTERNSHIP
}
```

```
}

enum Plan {
    STANDARD
    FEATURED
    FEATURE_UPGRADE
}

enum OrderStatus {
    PENDING
    PAID
    FAILED
    REFUNDED
}

enum SubStatus {
    PENDING
    CONFIRMED
    UNSUBSCRIBED
}

enum Frequency {
    DAILY
    WEEKLY
}

enum IssueStatus {
    DRAFT
    SENT
}

model Company {
    id          String    @id @default(cuid())
    slug        String    @unique
    name        String
    websiteUrl  String
    domain      String    // e.g. northwindsecurity.com (used for claim check)
    logoUrl     String?
    description String?
    blocklisted Boolean @default(false)
    createdAt   DateTime @default(now())
    jobs        Job[]
}
```

```

    sources      AtsSource[]
    claims       CompanyClaim[]
}

model AtsSource {
  id             String      @id @default(cuid())
  companyId      String
  company        Company     @relation(fields: [companyId], references: [id])
  provider       AtsProvider
  boardToken     String      // greenhouse token / lever company / ashby board
  active         Boolean     @default(true)
  lastSyncAt     DateTime?
  failedRuns     Int         @default(0)
  jobs           Job[]
  logs           SyncLog[]
  @@unique([provider, boardToken])
}

model Job {
  id             String      @id @default(cuid())
  slug           String      @unique
  origin         JobOrigin
  status         JobStatus
  companyId      String
  company        Company     @relation(fields: [companyId], references: [id])

  title          String
  descriptionHtml String
  category       String
  seniority      String?
  employmentType EmploymentType
  locationType   LocationType
  eligibleRegions String[]    // ["WORLDWIDE"] or country codes / region codes
  locationText   String?     // for hybrid/on-site
  salaryMin      Int?
  salaryMax      Int?
  salaryCurrency String?
  salaryPeriod   String?     // YEAR | MONTH | HOUR
  applyUrl       String?
  applyEmail     String?

  // ATS import fields

```

```

atsSourceId      String?
atsSource        AtsSource?    @relation(fields: [atsSourceId], references: [id]
sourceJobId      String?
contentHash      String?
lastSeenAt       DateTime?
missedRuns       Int           @default(0)

// Partner feed fields (origin = FEED). applyUrl holds the partner's tracking li
feedPartnerId    String?
feedPartner      FeedPartner?  @relation(fields: [feedPartnerId], references: [i

// Direct posting fields
employerEmail    String?
editToken        String?       @unique
approvedBefore   Boolean       @default(false)

// Lifecycle and featuring
isFeatured       Boolean       @default(false)
featuredUntil    DateTime?
publishedAt      DateTime?
expiresAt        DateTime?
indexNotifiedAt  DateTime?      // last time Google Indexing API was noti
createdAt        DateTime       @default(now())
updatedAt        DateTime       @updatedAt

orders           Order[]
clicks           JobClick[]
dailyStats       JobDailyStat[]

@@unique([atsSourceId, sourceJobId])
@@unique([feedPartnerId, sourceJobId])
@@index([status, isFeatured, publishedAt])
@@index([companyId])
}

// Add a generated tsvector column + GIN index for full-text search via a raw SQL

model Order {
  id            String          @id @default(cuid())
  jobId        String
  job           Job             @relation(fields: [jobId], references: [id])
  plan         Plan
  provider      String          // "stripe" | "lemonsqueezy"

```

```

providerSessionId String?    @unique
amountCents      Int
currency         String      @default("USD")
feeCents         Int?
status           OrderStatus @default(PENDING)
payerEmail       String
receiptUrl       String?
createdAt        DateTime     @default(now())
paidAt           DateTime?
}

model WebhookEvent {          // idempotency
  id          String  @id @default(cuid())
  provider    String
  eventId     String
  receivedAt  DateTime @default(now())
  @@unique([provider, eventId])
}

model JobClick {              // one row per apply click
  id          String  @id @default(cuid())
  jobId       String
  job         Job      @relation(fields: [jobId], references: [id])
  at          DateTime @default(now())
  referrer    String?
  @@index([jobId, at])
}

model JobDailyStat {          // page views counter per day
  jobId       String
  job         Job      @relation(fields: [jobId], references: [id])
  date        DateTime @db.Date
  views       Int       @default(0)
  applyClicks Int       @default(0)
  @@id([jobId, date])
}

model CompanyClaim {
  id          String  @id @default(cuid())
  companyId   String
  company     Company @relation(fields: [companyId], references: [id])
  email       String

```

```

    tokenHash String
    expiresAt DateTime
    verifiedAt DateTime?
    createdAt DateTime @default(now())
}

model Subscriber {
  id String @id @default(cuid())
  email String @unique
  status SubStatus @default(PENDING)
  frequency Frequency @default(WEEKLY)
  categories String[]
  regions String[]
  confirmToken String @unique
  unsubscribeToken String @unique
  createdAt DateTime @default(now())
  confirmedAt DateTime?
}

model NewsletterIssue {
  id String @id @default(cuid())
  subject String
  bodyHtml String
  status IssueStatus @default(DRAFT)
  createdAt DateTime @default(now())
  sentAt DateTime?
}

model SyncLog {
  id String @id @default(cuid())
  sourceId String
  source AtsSource @relation(fields: [sourceId], references: [id])
  startedAt DateTime @default(now())
  finishedAt DateTime?
  seen Int @default(0)
  added Int @default(0)
  updated Int @default(0)
  expired Int @default(0)
  error String?
}

enum SponsorStatus {

```

```

INQUIRY
PAID
SCHEDULED
SENT
CANCELLED
}

model FeedPartner {
  id          String    @id @default(cuid())
  name        String    @unique           // e.g. "JobNet"
  feedUrl     String
  active       Boolean   @default(true)
  maxJobs      Int       @default(500)
  estCpcCents  Int?      // estimate for admin reports only
  lastSyncAt   DateTime?
  jobs         Job[]
}

model AffiliateOffer {
  id          String    @id @default(cuid())
  name        String
  url          String    // affiliate link
  category     String    // RESUME | COURSE | CERTIFICATION |
  headline     String
  body         String
  placements   String[]  // JOB_PAGE | ALERT_EMAIL | NEWSLETTER
  active       Boolean   @default(true)
  clicks       AffiliateClick[]
  createdAt    DateTime @default(now())
}

model AffiliateClick {
  id          String    @id @default(cuid())
  offerId     String
  offer       AffiliateOffer @relation(fields: [offerId], references: [id])
  placement   String
  at          DateTime   @default(now())
  @@index([offerId, at])
}

model SponsorBooking {
  id          String    @id @default(cuid())

```

```

company      String
email        String
linkUrl      String
headline     String
bodyText     String
logoUrl      String?
priceCents   Int
status       SponsorStatus @default(INQUIRY)
issueId      String?
createdAt    DateTime       @default(now())
}

model Takedown {
  id          String @id @default(cuid())
  url         String
  requesterEmail String
  reason      String
  status      String @default("OPEN") // OPEN | DONE | REJECTED
  createdAt    DateTime @default(now())
}

```

*Note for the AI:* the schema above targets Prisma 6 syntax. If you use a newer Prisma major version, follow its current docs for datasource/config setup (for example a separate Prisma config file) and keep the models the same.

### Status transitions for `Job.status`:

```

DIRECT: PENDING_PAYMENT -> (paid) -> PENDING_REVIEW -> (approve) -> LIVE -> (30
days) -> EXPIRED
                                           \-> LIVE (trusted employer, auto)      \->
(renew) -> LIVE
      PENDING_REVIEW -> (reject) -> REJECTED (+ refund)
ATS:   (import) -> LIVE -> (missing 2 runs) -> EXPIRED ; any -> REMOVED
      (takedown / admin)
FEED:  (import) -> LIVE -> (missing 2 runs) -> EXPIRED ; partner switched off -
> EXPIRED

```

## 11. Automatic job import specification (ATS feeds)

### 11.1 Sources (public, no login)

Verify exact parameters against each provider's current official documentation before coding.

| Provider   | Public endpoint pattern                                                                                              | Notes                                                                                                      |
|------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Greenhouse | <code>GET https://boards-api.greenhouse.io/v1/boards/{boardToken}/jobs?content=true&amp;pay_transparency=true</code> | Board token is the last part of <code>boards.greenhouse.io/{token}</code> .<br>No API key for reading jobs |
| Lever      | <code>GET https://api.lever.co/v0/postings/{company}?mode=json</code> (EU accounts: <code>api.eu.lever.co</code> )   | Company name is the last part of <code>jobs.lever.co/{company}</code>                                      |
| Ashby      | <code>GET https://api.ashbyhq.com/posting-api/job-board/{boardName}?includeCompensation=true</code>                  | Skip postings where <code>isListed</code> is false                                                         |

## 11.2 Normalization (map every provider to the `Job` model)

| Job field       | Greenhouse                                          | Lever                                           | Ashby                                              |
|-----------------|-----------------------------------------------------|-------------------------------------------------|----------------------------------------------------|
| sourceJobId     | <code>id</code>                                     | <code>id</code>                                 | <code>id</code>                                    |
| title           | <code>title</code>                                  | <code>text</code>                               | <code>title</code>                                 |
| descriptionHtml | <code>content</code> (HTML, unescape then sanitize) | <code>description</code> + <code>lists</code>   | <code>descriptionHtml</code>                       |
| applyUrl        | <code>absolute_url</code>                           | <code>applyUrl</code> or <code>hostedUrl</code> | <code>applyUrl</code> or <code>jobUrl</code>       |
| location text   | <code>location.name</code>                          | <code>categories.location</code>                | <code>location</code>                              |
| remote flag     | parse location text / metadata                      | <code>workplaceType</code>                      | <code>isRemote</code> / <code>workplaceType</code> |
| employment type | metadata / default FULL_TIME                        | <code>categories.commitment</code>              | <code>employmentType</code>                        |
| salary          | <code>pay_input_ranges</code> (if present)          | <code>salaryRange</code> (if present)           | <code>compensation</code> (if present)             |
| published date  | <code>first_published</code>                        | <code>createdAt</code>                          | <code>publishedAt</code>                           |

## 11.3 Processing steps (per source, per night)

1. Fetch with 10 s timeout, max 1 request per second per host, 2 retries with backoff. Send a clear `User-Agent` with a contact email.

2. Normalize (11.2). Strip personal data (no recruiter names or emails).
3. **Sanitize HTML** with an allow-list (p, ul, ol, li, strong, em, a, h2, h3, br). Remove scripts, styles, iframes, inline event handlers.
4. **Niche filter:** keep the job only if the title (or category) matches `includeKeywords` and does not match `excludeKeywords`. Assign one category from `NICHE.categories` by keyword rules; unmatched jobs are skipped.
5. **Eligibility:** if `locationType = REMOTE`, try to parse eligible countries/regions from the location text. If unknown, set `eligibleRegions = []` (job is shown as "Remote — see description" and gets **no JSON-LD**, see Section 12).
6. **Upsert** by `(atsSourceId, sourceJobId)`. Compute `contentHash` from title + description + location + salary. If the hash changed → update fields and set `updatedAt`.
7. Set `lastSeenAt = now`, `missedRuns = 0`.
8. After the loop, every job of this source that was **not** in the feed: `missedRuns += 1`. If `missedRuns >= 2` → status `EXPIRED`.
9. **Duplicate check:** if a DIRECT job of the same company has the same normalized title and location, hide the ATS copy (status `REMOVED` with note "duplicate").
10. Save a `SyncLog` row. If the fetch failed, increase `AtsSource.failedRuns`; after 3 failed nights email the Owner.

#### 11.4 Rules to stay respectful and safe

- Only import jobs that the employer has published publicly. Skip unlisted ones.
- Always send the candidate to the employer's own apply URL. Never copy candidate data.
- Show the label "**Direct from company career page**" on imported jobs.
- Honor takedown requests within 48 hours and blocklist the company source.
- Never overwrite an imported job's text by hand; the employer's system is the source of truth.
- Respect rate limits; back off on HTTP 429/5xx.

#### 11.5 Partner feed import (pay-per-click backfill)

- Partner networks provide an XML (or JSON) feed URL after approval. Read the partner's feed documentation for field names.
- Map each feed item to the `Job` model (title, description, company name, location, remote flag if present, apply/tracking URL). If `employmentType` is missing, default to `FULL_TIME`; if `locationType` is missing, import the job only if the location text or keywords indicate remote work.
- Apply the **same niche filter** (Section 11.3, step 4) and the same HTML sanitizer.
- Save the partner's tracking link in `applyUrl` **unchanged**; the click route redirects to it.

- Upsert by `((feedPartnerId, sourceJobId))`; use a hash of the link if the feed has no id. Expire after 2 missed runs. Cap per partner `((maxJobs))`.
  - Sort so that partner jobs are shown **after** DIRECT and ATS jobs, in their own "Sponsored" section.
  - Log each run in a simple `((SyncLog))`-style record (add a nullable `((feedPartnerId))` if needed) and alert the Owner when a feed fails 3 nights in a row.
- 

## 12. SEO and Google Jobs specification

### 12.1 Job page structured data (`((JobPosting))` JSON-LD)

Render one `<script type="application/ld+json">` **only on** `((/jobs/[slug]))` and **only when status is LIVE**.

```
json
```

```

{
  "@context": "https://schema.org",
  "@type": "JobPosting",
  "title": "SOC Analyst (Remote)",
  "description": "<p>Full HTML description...</p>",
  "datePosted": "2026-09-20",
  "validThrough": "2026-10-20T23:59:59Z",
  "employmentType": "FULL_TIME",
  "hiringOrganization": {
    "@type": "Organization",
    "name": "Acme Cloud",
    "sameAs": "https://acmecloud.example",
    "logo": "https://.../logo.png"
  },
  "jobLocationType": "TELECOMMUTE",
  "applicantLocationRequirements": [
    { "@type": "Country", "name": "IN" },
    { "@type": "Country", "name": "PH" }
  ],
  "baseSalary": {
    "@type": "MonetaryAmount",
    "currency": "USD",
    "value": { "@type": "QuantitativeValue", "minValue": 40000, "maxValue": 60000,
  },
  "identifier": { "@type": "PropertyValue", "name": "NicheJobs", "value": "x7k2" }
  "directApply": false
}

```

## 12.2 Rules (Google's job posting guidelines — follow strictly)

1. Put JobPosting markup **only on the page of a single job**, never on list, search or category pages.
2. Each job page must let the user apply (a visible apply button/link on that page). A page with no way to apply causes a policy violation.
3. `hiringOrganization` must be the **real employer**, not NicheJobs.
4. Remote jobs need `jobLocationType: "TELECOMMUTE"` and `applicantLocationRequirements`. If eligible regions are unknown, output **no JSON-LD** for that job (it can still be listed on the site).
5. Include `validThrough`. Remove the markup as soon as the job is expired or removed.
6. Description must contain the real job text (no keyword stuffing).

7. Validate samples with Google's Rich Results Test before launch.
8. Partner-feed (`(FEED)`) jobs: **no** JobPosting markup, (`noindex, follow`), not in the sitemap, and never sent to the Indexing API. They exist for user value and click income, not for search traffic.

## 12.3 Indexing API

- Use the Google Indexing API **only for URLs with JobPosting markup**.
- Send (`URL_UPDATED`) when a job goes LIVE or changes; (`URL_DELETED`) when it expires or is removed.
- Default quota is about 200 requests/day. Add (`indexNotifiedAt DateTime?`) to (`Job`). The cron picks jobs where (`indexNotifiedAt IS NULL OR updatedAt > indexNotifiedAt`), **direct paid jobs first**, max 180 per day.
- Also keep (`sitemap.xml`) (with (`lastmod`)) and verify the domain in Google Search Console.

## 12.4 Other SEO requirements

- Unique (`<title>`) and meta description per page, canonical URL, Open Graph image (auto-generated card with job title + company).
- Clean stable URLs. Job slug never changes after publish.
- **Programmatic landing pages** (`(/remote-[category]-jobs)`, `(/remote-jobs-in-[country])`): only indexable if they have **at least 5 LIVE direct or ATS jobs** (partner-feed jobs do not count; otherwise (`noindex`)). Each has an original intro paragraph and shows the job list. Never duplicate the same intro text across pages.
- Internal links: footer with categories and countries; "similar jobs" on each job page; company page ↔ job page links.
- Performance: Core Web Vitals green, images lazy-loaded, no layout shift.
- Add value beyond copied text: normalized salary, clear region eligibility, similar jobs, company info.

---

## 13. Payments specification

### 13.1 Provider interface (keep providers swappable)

ts

```

export type PaymentEvent = {
  eventId: string;
  type: "PAID" | "FAILED" | "REFUNDED";
  providerSessionId: string;
  amountCents: number;
  currency: string;
  feeCents?: number;
  receiptUrl?: string;
};

export interface PaymentProvider {
  createCheckout(input: {
    orderId: string; jobId: string; plan: "STANDARD" | "FEATURED" | "FEATURE_UPGRADE";
    amountCents: number; currency: string; payerEmail: string;
    successUrl: string; cancelUrl: string;
  }): Promise<{ url: string; providerSessionId: string }>;
  verifyWebhook(req: Request): Promise<PaymentEvent | null>; // verifies signature
  refund(providerSessionId: string): Promise<void>;
}

```

## 13.2 Rules

1. **Price is decided on the server** from `config/pricing.ts`. Never trust a price sent by the browser.
2. A job becomes LIVE / PENDING\_REVIEW **only from a verified webhook**, never from the success-page redirect.
3. Verify the webhook signature, check `amountCents` and `currency` match the Order, and record the event in `WebhookEvent` (unique) so duplicates are ignored.
4. Put `orderId` in provider metadata so the webhook can find the Order.
5. Use the provider's **hosted checkout page** (no card data touches our server).
6. Store `feeCents` when the provider reports it, for revenue reports.
7. Taxes: a merchant-of-record provider (for example Lemon Squeezy or Paddle) collects and pays sales tax/VAT for the sale. With a direct gateway (for example Stripe), the Owner is responsible for tax rules — the Owner should ask an accountant.
8. Test everything in the provider's test mode first (successful, failed, refunded, duplicate webhook).

### 13.3 Provider notes for the Owner (not for the AI)

- Stripe availability depends on the country. In India, Stripe is invite-only and requires a registered business, not an individual.
  - Merchant-of-record providers (Lemon Squeezy, Paddle) accept sales globally and pay out to a bank; confirm that selling job listings is an allowed product category for the chosen provider.
  - Razorpay International is an option for a registered Indian business (fees and tax handling differ; it does not handle foreign taxes for you).
- 

### 14. Email specification

Sender: NicheJobs <hello@yourdomain> with SPF, DKIM and DMARC set up. Every email has a plain-text version.

| #  | Email                     | To              | Trigger                               | Subject (example)                               |
|----|---------------------------|-----------------|---------------------------------------|-------------------------------------------------|
| 1  | Confirm subscription      | Candidate       | Signup                                | "Confirm your job alerts"                       |
| 2  | Job alert digest          | Subscriber      | Daily/weekly cron                     | "6 new remote security jobs"                    |
| 3  | Weekly newsletter         | Subscribers     | Owner clicks Send                     | "This week: 12 remote security jobs"            |
| 4  | Listing received          | Employer        | Order PAID                            | "We received your job listing"                  |
| 5  | Listing live              | Employer        | Approved / auto-live                  | "Your job is live on NicheJobs"                 |
| 6  | Listing rejected + refund | Employer        | Owner rejects                         | "We couldn't publish your listing"              |
| 7  | Finish your listing       | Employer        | 1 h and 24 h after abandoned checkout | "Your job draft is waiting"                     |
| 8  | Renewal reminder          | Employer        | Day 25                                | "Your listing expires in 5 days"                |
| 9  | Listing expired           | Employer        | Day 30                                | "Your listing has expired — renew in one click" |
| 10 | Claim magic link          | Company contact | Claim started                         | "Verify your company on NicheJobs"              |
| 11 | New listing to review     | Owner           | Order PAID (new employer)             | "Review: SOC Analyst at Acme Cloud"             |
| 12 | Import failing            | Owner           | 3 failed nights                       | "Source failing: northwindsecurity"             |

### Example — email 5 "Your job is live":

Hi Maria, your job **SOC Analyst (Remote)** is now live: {public link}. It stays online until {expiry date}. Track views and apply clicks here: {manage link}. Want more visibility? Upgrade to Featured for the difference and get top placement and the newsletter slot.

**Rules:** unsubscribe link and one-click unsubscribe header on all alert/newsletter emails; double opt-in for subscribers; rate-limit sending; log all sends; never email candidates about anything but jobs they asked for. Each alert or newsletter may contain **at most one sponsor**

**block and one affiliate block**, placed after the jobs and clearly labeled **Sponsored / Partner offer**.

---

## 15. UI/UX design brief (for the designer AI)

### 15.1 Feel

Fast, clean, trustworthy, and calm. Think "serious tool for professionals", not "flashy startup". The job list is the hero. Dense but readable rows. One accent color. Lots of white space around content, not inside the list rows.

### 15.2 Design tokens (DEFAULT — adjust to the niche)

- **Colors:** neutral grays for text/background, one accent color (for buttons and links), a warm highlight color for **Featured** jobs (soft background + small "Featured" badge), success green, warning amber, error red. Support light and dark mode.
- **Typography:** one clean sans-serif family (for example Inter). Sizes: 14–16 px body, 20–28 px page titles, 32–44 px hero headline. Comfortable line height (1.5–1.7 for job descriptions).
- **Spacing:** 4-px scale. Cards with 12–16 px radius. Subtle borders, very light shadows.
- **Iconography:** simple line icons (lucide).

### 15.3 Key components

- `JobCard` (list row): company logo (or letter avatar), title, company, chips (Remote region, employment type, salary), posted date, badges (Featured, Direct from company). Whole row clickable; Apply button optional on desktop.
- `FilterBar` / sidebar with filters; on mobile it opens as a bottom sheet.
- `SalaryChip`, `RemoteChip`, `FeaturedBadge`, `DirectBadge`.
- `PostJobStepper` (3 steps) with live preview panel.
- `PricingCards` (Standard, Featured) with clear differences.
- `StatsChart` (views and apply clicks per day) on manage page.
- `EmptyState`, `Skeleton` loaders, `Toast` messages, `ConfirmDialog`.

### 15.4 States that must be designed

Loading (skeletons), empty results ("No jobs match — try removing a filter" + alert signup), form errors (inline), payment cancelled, payment pending, expired job page, 404, offline/error.

### 15.5 Responsive and accessibility

- Mobile-first. Test at 360 px, 768 px, 1280 px.
- Keyboard accessible, visible focus rings, color contrast AA, proper labels on all form fields, alt text for logos.

- No horizontal scrolling on mobile.

### **15.6 Performance targets**

Largest Contentful Paint under 2.5 s on mid-range mobile; total JS on job page as small as possible; images optimized; fonts self-hosted.

### **15.7 Tone of voice (copy)**

Short, plain, helpful. Examples:

- Hero: "Remote cybersecurity jobs, direct from companies."
  - Post-a-job: "Reach security professionals in minutes. No account needed."
  - Empty state: "Nothing matches yet. Get an email when a new job appears."
- 

## **16. Admin specification (owner only)**

**Access:** Magic-link login. Only emails listed in `ADMIN_EMAILS` can sign in. Session cookie, 7 days. Every admin route checks the session.

| Page                           | What the Owner can do                                                                                                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/admin</code>            | See KPIs: revenue (this month / last month), paid listings, live jobs, imported vs direct, visitors, apply clicks, subscribers, sync health                                  |
| <code>/admin/review</code>     | Queue of <code>PENDING_REVIEW</code> . Shows job preview, company website link, email domain, red-flag warnings. Buttons: Approve, Reject (reason required → email + refund) |
| <code>/admin/jobs</code>       | Search all jobs, change status, feature/unfeature, extend expiry, delete/remove                                                                                              |
| <code>/admin/companies</code>  | Companies with stats (jobs, views, apply clicks last 30 days), claim status, blocklist toggle, "copy outreach email" helper                                                  |
| <code>/admin/sources</code>    | Add/edit/pause ATS sources, see last sync result, run sync now, failures                                                                                                     |
| <code>/admin/partners</code>   | Add/edit/pause partner feeds, run sync, clicks per partner and estimated income vs. real payout (entered manually)                                                           |
| <code>/admin/affiliates</code> | Manage affiliate offers, see clicks per offer/placement, record commissions received                                                                                         |
| <code>/admin/sponsors</code>   | Sponsor bookings: inquiry → paid → scheduled → sent; attach to newsletter issue                                                                                              |
| <code>/admin/orders</code>     | List of orders, refunds, totals, export CSV                                                                                                                                  |
| <code>/admin/newsletter</code> | Draft, edit, preview, send test, send                                                                                                                                        |
| <code>/admin/takedowns</code>  | Handle removal requests                                                                                                                                                      |
| <code>/admin/settings</code>   | Prices, niche keywords/categories (read-only view in MVP), site texts                                                                                                        |

**Red-flag rules shown in the review queue:** free email domain; email domain differs from website domain; description mentions fees paid by candidate, crypto, gift cards, or "WhatsApp only"; no company website; salary unrealistic (top 1% of category); duplicate of another live job.

## 17. Security, privacy and legal

### 17.1 Security checklist

- Validate all inputs on the server (zod). Sanitize all HTML from employers and imports (allow-list).

- Edit tokens: 32+ random bytes, stored hashed if possible, never shown in public pages, single job scope.
- Cron and webhook endpoints are protected (secret header / signature).
- Rate limiting on: post-a-job, subscribe, claim, takedown, click redirect.
- Turnstile CAPTCHA + honeypot on public forms.
- Apply redirect: allow only `(http/https)` URLs; block `(javascript:)` and private IP ranges.
- Admin routes protected; no admin data in client bundles.
- Secrets only in environment variables. Logs never contain tokens or emails in plain text where avoidable.

## 17.2 Privacy

- NicheJobs stores almost no candidate data: optional alert email + preferences only.
- Apply happens on the employer's site.
- Analytics is privacy-friendly and cookie-less if possible (then no cookie banner is needed; check local law).
- Privacy page explains: what is stored, why, how to unsubscribe and delete data (email request within 30 days).

## 17.3 Legal pages (must exist before launch)

Terms of Service (employer rules: real jobs only, no fees charged to candidates, no discrimination, refund policy), Privacy Policy, Takedown/Contact page, Imprint/company details as required in the Owner's country. The Owner should have these reviewed by a professional; the AI must not present them as legal advice.

## 17.4 Content rules for listings

Allowed: real, current jobs in the niche, with an apply method. Not allowed: scams, pay-to-apply, multi-level marketing, misleading salary, discriminatory requirements, jobs outside the niche (Owner may reject).

## 17.5 Labeling and disclosure (money-related content)

- Partner jobs and sponsor blocks are labeled **Sponsored**. Affiliate blocks show "We may earn a commission if you buy."
  - All partner/affiliate/sponsor links use `(rel="sponsored nofollow")`.
  - Follow each partner's publisher terms; the Owner is responsible for tax and legal rules on income in his country and for affiliate/advertising disclosure laws where the visitors live.
-

## 18. Analytics and KPIs

### 18.1 Events to track

`job_view`, `apply_click`, `post_job_started`, `post_job_step_completed`, `checkout_started`,  
`order_paid`, `alert_subscribed`, `alert_confirmed`, `newsletter_sent`, `claim_started`,  
`claim_verified`, `feature_upgrade_paid`.

### 18.2 Weekly KPIs (shown in `/admin`)

| KPI                                                                                | Why                                                 |
|------------------------------------------------------------------------------------|-----------------------------------------------------|
| Live jobs (imported / direct)                                                      | Board is full enough to be useful                   |
| Visitors, job views                                                                | Traffic                                             |
| Apply clicks and apply clicks per job                                              | Proof of value to employers                         |
| Subscribers (confirmed)                                                            | Owned audience                                      |
| Partner clicks per 1,000 visitors and estimated income per click                   | Bridge income (Stream 1)                            |
| Affiliate clicks and commissions                                                   | Stream 2                                            |
| Sponsor slots sold and sponsor inquiries                                           | Stream 5                                            |
| <b>Revenue by stream</b> (partner, affiliate, listings, featured, sponsors, plans) | Shows concentration risk and what to double down on |
| Employers contacted → claimed → paid                                               | Sales funnel                                        |
| Paid listings, revenue, net revenue                                                | Money                                               |
| Checkout conversion (started → paid)                                               | Pricing and form quality                            |
| Renewal rate                                                                       | Retention                                           |

### 18.3 Example targets (starting guesses, adjust with real data)

- Launch: 200+ live jobs from 100–200 company sources.
- First paid listing: within 30–60 days of launch.
- Track visitors, subscribers and apply clicks weekly and compare week over week; growth matters more than any single number. If the numbers stay flat for several weeks, change tactics (Section 19) before changing the code.

## 19. Go-to-market plan (how people find the site and how employers start paying)

### 19.1 Step 0 — choose the niche before coding (Owner task, 3 days)

A niche is good only if it passes all three tests:

1. **Hiring is constant.** On LinkedIn Jobs, filter the niche to the last 30 days. 200+ active postings = a real market. About 20 = keep looking.
2. **Employers have budget and use modern career systems.** Check 30 companies in the niche: do they post on Greenhouse, Lever or Ashby? (This is what fills your board on day one.)
3. **No strong niche board already owns Google page one** for the niche keywords. General boards (Indeed, LinkedIn) are not your competition; other niche boards are.

Niches to test (Owner's shortlist, unverified):

- Cybersecurity, healthcare or engineering roles — harder-to-fill, higher-salary fields tend to support higher listing prices.
- Remote roles open to candidates in a specific region (for example India/APAC time zones) — ATS feeds already carry location and remote fields.
- Avoid niches that are too broad ("remote jobs") or too dead (rarely hiring), and avoid niches already well served (for example plain "Next.js jobs").

## 19.2 Timeline

| When           | Owner does                                                                                                                       | AI/app does                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Days 1-3       | Niche test (19.1). Pick name + domain                                                                                            | —                                            |
| Weeks 1-5      | Build with the AI in milestones (Section 21). Meanwhile collect 100-200 company career links                                     | Milestones M1-M8                             |
| Week 3-5       | <b>Apply to 2-3 partner job networks and 2-3 affiliate programs</b> (approval takes time). Note 2-3 possible sponsors            | Partner feed and affiliate blocks ready (M7) |
| Week 4-5       | <b>Email 10-20 target employers before launch</b> ("we will list your roles for free, would you like early featured placement?") | Sources added, import runs                   |
| Week 5         | <b>Launch:</b> post on LinkedIn, X, relevant Reddit/Discord/Slack communities, Indie Hackers, Product Hunt                       | —                                            |
| Weeks 6-10     | 15-20 personalized outreach emails per weekday (Flow E). Publish the weekly newsletter. Share salary/insight posts               | Cron jobs run                                |
| Week 10 (gate) | If there is <b>no paid listing</b> and traffic is flat → change the niche (Section 20)                                           | —                                            |

## 19.3 Traffic channels (free first)

1. **Google Jobs / Google Search** — correct JobPosting markup + landing pages (biggest long-term source).
2. **Newsletter** — every visitor is offered alerts; weekly issue drives repeat visits.
3. **Communities in the niche** — LinkedIn groups, subreddits, Discords, Telegram/WhatsApp groups. Share useful posts (salary insights, "jobs open to [region] this week"), not spam.
4. **Founder posts** — build in public on X/LinkedIn.
5. **Product Hunt / Indie Hackers launch.**

## 19.4 Who pays, and why they say yes

- The employer already has jobs on the board (Flow D). The Owner shows real numbers ("96 apply clicks in 30 days").
- The offer is simple: "Featured placement for \$99: top of list for 30 days + newsletter slot."
- Employers with steady hiring (agencies, startups with funding, high-turnover teams) are the best targets.

- Typical objection: "My jobs are already listed." Answer: "Yes, free. Featured gets ~N times more views on our board" (use real numbers once available).

## 19.5 Outreach email templates (Owner sends manually — do not automate mass email)

### Template 1 — Imported jobs, upgrade offer

Subject: Your 3 security roles got 96 apply clicks on NicheJobs Hi {first name}, I run NicheJobs, a job board for remote cybersecurity roles. Your open roles ({role 1}, {role 2}) are already listed for free and got {N} views and {M} apply clicks in the last 30 days. If you'd like them at the top of the list plus in our weekly newsletter, Featured placement is \$99 for 30 days. You can claim your company page here: {claim link}. If you'd rather not be listed, reply "remove" and I'll take it down today. — {Owner name}, {address/website}

### Template 2 — Pre-launch, no jobs yet

Subject: Free early listing for {company}'s security roles Hi {first name}, I'm launching NicheJobs, a curated board for remote cybersecurity roles. I'd like to list {company}'s open roles at launch, free, with a link to your careers page. Would that be OK? Early employers also get first choice of featured spots.

**Legal note for outreach:** send only to business contacts, identify yourself, include a real address/opt-out, and follow the anti-spam and privacy rules of the countries you email (for example CAN-SPAM, GDPR/PECR). The Owner is responsible for compliance.

## 19.6 Employer FAQ (put on /employers)

- How long does my job stay live? 30 days.
- Do I need an account? No.
- Can I edit my job? Yes, with your private manage link.
- What if my job is rejected? Full refund.
- How will candidates apply? On your website or by email; we never collect CVs.
- Do you show salary? Yes when provided; jobs with salary get more clicks.
- Can I get an invoice? Yes, from the payment receipt link.

## 19.7 First-90-days money checklist (Owner)

1. **Before launch:** apply to partner networks and affiliate programs; list 10–20 employers to email; list 3 possible sponsors.
2. **Launch to day 30:** switch on partner feed + affiliate blocks as soon as approved; grow the newsletter list; send the weekly issue; contact 15–20 employers per weekday.
3. **Day 30–60:** first paid listings and feature upgrades; publish real numbers (views, apply clicks) in outreach; raise price if the first 10 sales come easily.

4. **Day 60-90:** offer the first sponsor slot when the newsletter has about 1,000+ confirmed subscribers; offer the monthly plan to employers who bought twice.
  5. **Every week:** check revenue by stream in `/admin`, drop what earns nothing, double down on what works.
- 

## 20. Roadmap and risks

### 20.1 After the MVP (in this order)

1. Employer monthly plans (about \$399/month) once about 20 employers have paid.
2. In-app checkout for sponsor slots; sell recurring sponsorships.
3. Banner ads (AdSense first, direct-sold ads when the audience is bigger).
4. Company profile pages with more content for paid employers.
5. Paid candidate tool (for example AI resume/cover-letter tailoring using the Claude API, about \$9/month benchmark) — treat as an experiment; candidates rarely pay.
6. Candidate profiles / resume database for recruiters (only after hundreds to thousands of profiles, with consent and deletion controls).
7. Virtual career fairs with sponsor tiers (only with a large, engaged audience).
8. More niches on the same codebase (config-driven).

## 20.2 Risks and how to handle them

| Risk                                                            | What to do                                                                                                                 |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Slow start (SEO takes months)                                   | Newsletter + community + direct outreach from day one; do not rely on Google alone                                         |
| Partner networks reject the site or per-click income is low     | Apply to several networks; treat Stream 1 as bridge income only; focus on employers and sponsors; grow traffic             |
| Depending on one revenue stream                                 | Track revenue by stream weekly; add the next stream on schedule (Section 5.3)                                              |
| Employers see their jobs listed free and do not pay             | Sell <b>Featured</b> + <b>newsletter</b> , show real click data, and offer direct-posting benefits (edit, stats, priority) |
| Wrong niche (too broad, too dead, too crowded)                  | Use the 3 tests (19.1) and the stop rule below                                                                             |
| Google policy problems (markup on wrong pages, no apply option) | Follow Section 12.2 exactly; test with Rich Results Test                                                                   |
| Copied content / duplicate text from ATS                        | Add value (salary, regions, similar jobs); link to the employer; honor takedowns                                           |
| Scam or fake listings                                           | Manual review of first listings, red-flag rules (Section 16), clear terms, quick removal                                   |
| Payment provider restrictions in Owner's country                | Keep provider adapters; decide provider before building checkout; confirm allowed category                                 |
| AI answers reducing search clicks                               | Job pages are transactional (apply), but do not depend on search only; build the newsletter and community audience         |
| One-person workload                                             | Automation (imports, expiry, reminders), 15-minute daily admin routine, simple features only                               |

## 20.3 Stop / change rule

If there is **no paid listing 10 weeks after launch** even after direct outreach, change the niche (the code stays; only `config/niche.ts` and the seed sources change).

## 21. Build milestones (for the AI model)

**M1 — Foundation.** Next.js + TypeScript + Tailwind project, Prisma schema (Section 10) and migrations, config files (Section 9.6), seed script with 30 sample jobs and 5 companies, base layout, admin magic-link login. *Done when:* app runs locally, seed loads, admin login works.

**M2 — Public site.** Home list, filters, full-text search, job page, company page, apply click redirect + `JobClick`, view counters, metadata, sitemap, robots, JSON-LD on job pages (Section 12). *Done when:* filters work together; job page passes Rich Results Test for a sample job; apply click is logged.

**M3 — Job import.** Greenhouse, Lever, Ashby adapters, normalization, sanitizer, niche filter, dedupe, expiry after 2 missed runs, `/admin/sources`, cron endpoint, `SyncLog`. *Done when:* adding a real board token imports matching jobs; removing a job upstream expires it after 2 runs.

**M4 — Employer posting and payment.** 3-step form with preview, logo upload, Turnstile, payment provider interface + Stripe adapter, webhook with idempotency, order + job statuses, transactional emails 4/5/7, `/manage/[token]` with stats, edit, close. *Done when:* test-mode payment creates the right statuses; duplicate webhook does nothing; manage link works.

**M5 — Review, expiry, Google.** `/admin/review` (approve/reject/refund), expiry + reminder cron (emails 8/9), Indexing API notifications, Lemon Squeezy adapter, renew and upgrade flows. *Done when:* approve makes job live and notifies Google; day-25/30 logic verified with time-travel tests.

**M6 — Audience and upgrades.** Alerts with double opt-in and unsubscribe, alert digest cron, newsletter draft/send, company claim (domain-verified magic link), company dashboard, Feature upgrade checkout for imported jobs. *Done when:* subscriber receives digest; claim works only with matching domain; upgrade sets `isFeatured`.

**M7 — Extra revenue streams.** Partner feed importer + `/admin/partners` + "Sponsored" partner section (Flow J, Section 11.5), affiliate offers + `/api/aff/[id]` + `/admin/affiliates` (Flow L), `/advertise` page + `SponsorBooking` + `/admin/sponsors` + sponsor block in newsletter (Flow K), revenue-by-stream KPIs in `/admin`. *Done when:* a test feed imports only niche jobs, partner jobs are `noindex` and labeled, partner and affiliate clicks are logged and redirect correctly, a sponsor booking appears in the next newsletter draft.

**M8 — Launch polish.** Programmatic landing pages (with 5-job rule), legal pages (including affiliate/sponsored disclosures), takedown form, analytics events, admin dashboard KPIs, accessibility and performance pass, error pages, deployment guide, production env checklist. *Done when:* Lighthouse mobile performance and accessibility scores are good, all pages pass the launch checklist (Appendix C).

**Definition of done for every milestone:** code compiles with no TypeScript errors, key flows have tests or a written manual test script, no secrets in the repo, README updated with run/deploy steps.

---

## Appendix A — Seed source file format (Owner fills this in)

data/sources.csv

```
company_name,website,provider,board_token
Northwind
Security,https://northwindsecurity.example,GREENHOUSE,northwindsecurity
Acme Cloud,https://acmecloud.example,ASHBY,acmecloud
Contoso Defense,https://contosodefense.example,LEVER,contosodefense
```

How to find a board token: open the company's careers page; if the apply links go to `boards.greenhouse.io/{token}`, `jobs.lever.co/{company}` or `jobs.ashbyhq.com/{name}`, the last part is the token.

## Appendix B — Manual test script (minimum)

1. Import a real board → jobs appear only if they match the niche.
2. Post a job with test payment → status PENDING\_REVIEW → approve → LIVE, email received, Google Indexing call logged.
3. Send the same webhook twice → only one state change.
4. Click Apply → click logged → redirected with `utm_source`.
5. Let a test job pass day 25 and 30 → reminder sent → expired, markup removed, deletion notice sent.
6. Subscribe → confirm → receive digest → unsubscribe in one click.
7. Claim a company with a matching email domain → success; with a different domain → rejected.
8. Submit a takedown request → visible in admin → job removed.

## Appendix C — Launch checklist

- ☐ Niche chosen, config filled, site name and domain set
- ☐ 100–200 sources added, 200+ jobs live
- ☐ Payment provider live mode tested with a real small payment and refund
- ☐ Applied to 2–3 partner job networks and 2–3 affiliate programs; partner feed tested (or switched off until approved)
- ☐ "Sponsored" labels and affiliate disclosure visible; links use `rel="sponsored nofollow"`

- ☐ `/advertise` page live with real audience numbers
- ☐ Legal pages published and reviewed
- ☐ Domain verified in Google Search Console, sitemap submitted
- ☐ Rich Results Test passed for 3 job pages
- ☐ Email domain: SPF, DKIM, DMARC pass
- ☐ Turnstile and rate limits active
- ☐ Backups enabled on database
- ☐ Admin login works; only owner emails allowed
- ☐ 10-20 employers contacted before launch

## Appendix D — Glossary

- **ATS:** Applicant Tracking System. Software companies use to post jobs (Greenhouse, Lever, Ashby).
- **ISR:** Incremental Static Regeneration — pages built ahead of time and refreshed in the background.
- **JSON-LD / JobPosting:** structured data that tells Google a page is a job.
- **Merchant of record:** a payment company that sells on your behalf and handles sales taxes.
- **Webhook:** a message the payment provider sends to your server when a payment happens.
- **Featured listing:** paid upgrade that pins a job at the top and adds it to the newsletter.
- **Magic link:** a login link sent by email, so no password is needed.